

Polynomial Regression

Machine Learning Assignment

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GitHub Repository: github.com/Sankalp-G2006/polynomial_regression

1 Introduction

This assignment focuses on predicting a continuous target variable y using polynomial regression on two personalized datasets, **var1** and **var2**. The datasets contain different numbers of input variables and have different maximum polynomial degrees.

For **var1**, six input variables $(x_1, \dots, x_6)$ are provided, with a maximum polynomial degree of 10. For **var2**, three input variables (x_1, x_2, x_3) are provided, with a maximum polynomial degree of 20.

The objective is to select an appropriate polynomial degree using cross-validation, train the final model using the available training data, and generate predictions for the provided test datasets.

The primary evaluation metrics specified for the assignment are Mean Squared Error (MSE) and coefficient of determination (R^2).

2 Methodology

The same general workflow is followed for both problems:

1. Load the corresponding training dataset and separate the input variables X from the target y .
2. Generate polynomial features up to the required degree.
3. Evaluate different polynomial degrees using 5-fold cross-validation.
4. Compare the validation MSE values.
5. Select the degree with the best validation performance among the evaluated degrees.
6. Train the final model using the complete training dataset.
7. Apply the trained model to the corresponding test dataset.
8. Generate the required prediction CSV file.

Polynomial features are generated using total polynomial degree. For example, for three variables and degree two, the feature expansion contains terms such as

$$1, x_1, x_2, x_3, x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2.$$

Thus, a polynomial of degree d contains terms for which the sum of the feature powers is at most d .

Regularization is also used where appropriate to control the effect of the large number of polynomial features, particularly at higher degrees.

3 Problem 1: var1

3.1 Dataset

The **var1** dataset contains six input variables,

$$x_1, x_2, x_3, x_4, x_5, x_6,$$

and the target variable y .

The training dataset contains 1000 samples and the test dataset contains 1000 samples. The target variable y is available in the training dataset but is not provided for the test dataset.

The maximum permitted polynomial degree for this problem is 10.

3.2 Degree Selection

Different polynomial degrees were evaluated using 5-fold cross-validation. Increasing the polynomial degree initially improves the model’s ability to capture nonlinear relationships. However, the number of polynomial features also increases rapidly, making regularization useful for controlling model complexity.

The degree-5 model was selected for the final **var1** model. Regularized regression was used to improve generalization in the expanded polynomial feature space.

Figure 1 shows the relationship between polynomial degree and validation error.

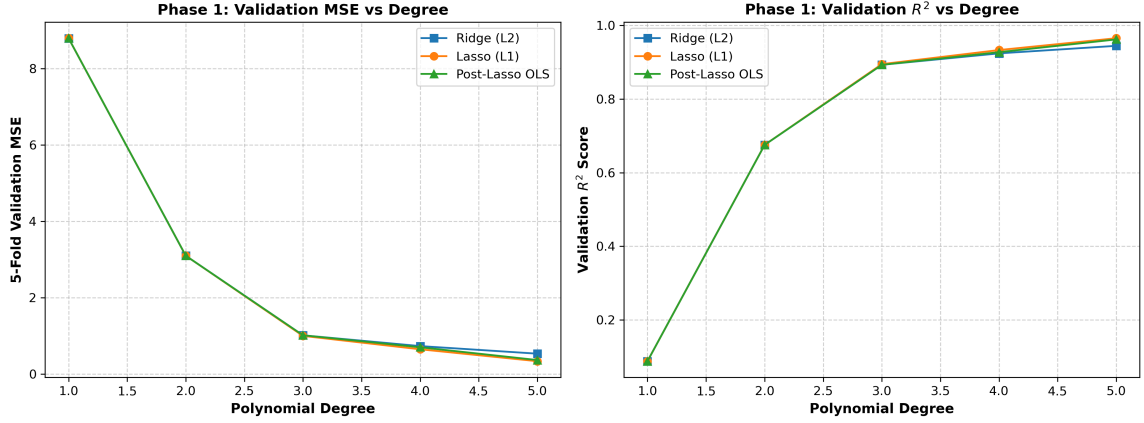


Figure 1: Validation error for the evaluated polynomial degrees for **var1**.

The degree-5 polynomial expansion produces 462 polynomial terms. After the feature-selection stage, 96 active terms are retained in the final model.

3.3 Final Model

The final **var1** model uses polynomial degree 5 with regularized regression and post-Lasso feature selection. The performance of the final model on the complete training dataset is:

Metric	Value
Training MSE	0.24041
Training R^2	0.97539

Table 1: Training performance of the final **var1** model.

These values describe performance on the training data and should not be interpreted as the final hidden-test evaluation.

3.4 Residual Analysis

Residuals provide an additional way of examining the behavior of the final regression model. Figure 2 shows the residual analysis for the final **var1** model.

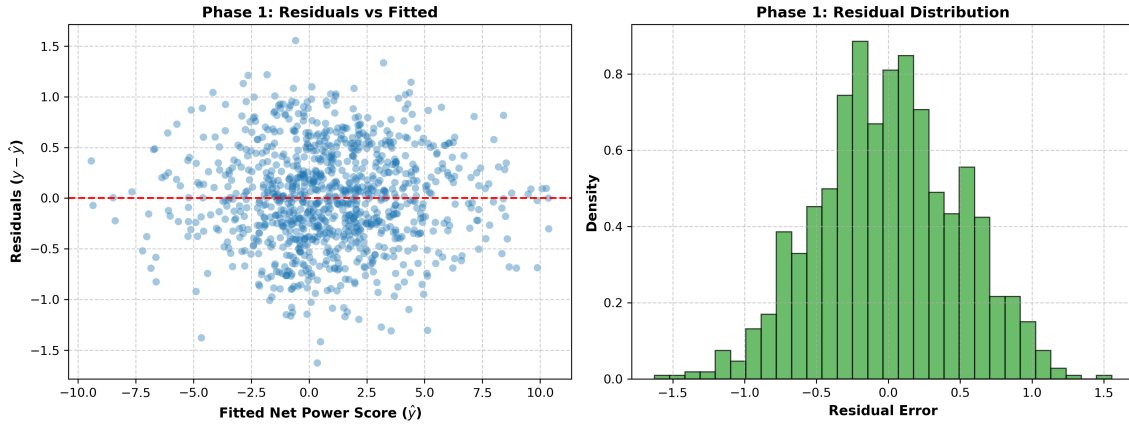


Figure 2: Residual analysis for the final `var1` model.

4 Problem 2: `var2`

4.1 Dataset

The `var2` dataset contains three input variables,

$$x_1, x_2, x_3,$$

and the target variable y .

The training dataset contains 1000 samples and the test dataset contains 1000 samples. The target variable is available only in the training dataset.

The maximum permitted polynomial degree for this problem is 20.

4.2 Degree Selection

Several polynomial degrees were evaluated using 5-fold cross-validation. Ridge regression with $\alpha = 0.05$ was used during the degree-selection experiment to control the effect of the increasing number of polynomial features.

The validation results are shown below.

Polynomial Degree	Ridge CV MSE
1	35.47803
2	21.15733
4	3.42043
6	0.58281
8	0.25566
10	0.24532
12	0.24759

Table 2: 5-fold cross-validation results for `var2`.

Figure 3 provides the corresponding visual comparison of validation error.

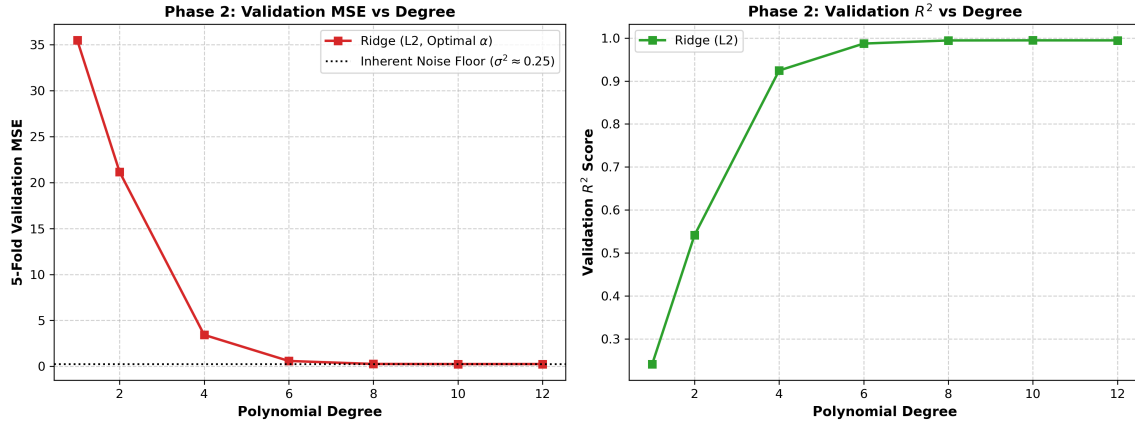


Figure 3: Validation MSE for the evaluated polynomial degrees for `var2`.

Degree 10 achieves the lowest validation MSE among the evaluated degrees. Therefore, degree 10 is selected for the final `var2` model.

4.3 Final Model

The final `var2` model uses:

- Polynomial degree: 10
- Ridge regression
- Regularization parameter: $\alpha = 0.05$

The degree-10 polynomial expansion contains 286 terms.

The cross-validation and full-training performance of the final model are:

Metric	Value
5-fold CV MSE	0.24532
5-fold CV R^2	0.99462
Full-training MSE	0.16666
Full-training R^2	0.99644
Full-training MAE	0.32906

Table 3: Performance of the final `var2` model.

4.4 Residual Analysis

Figure 4 shows the residual behavior of the final degree-10 Ridge model.

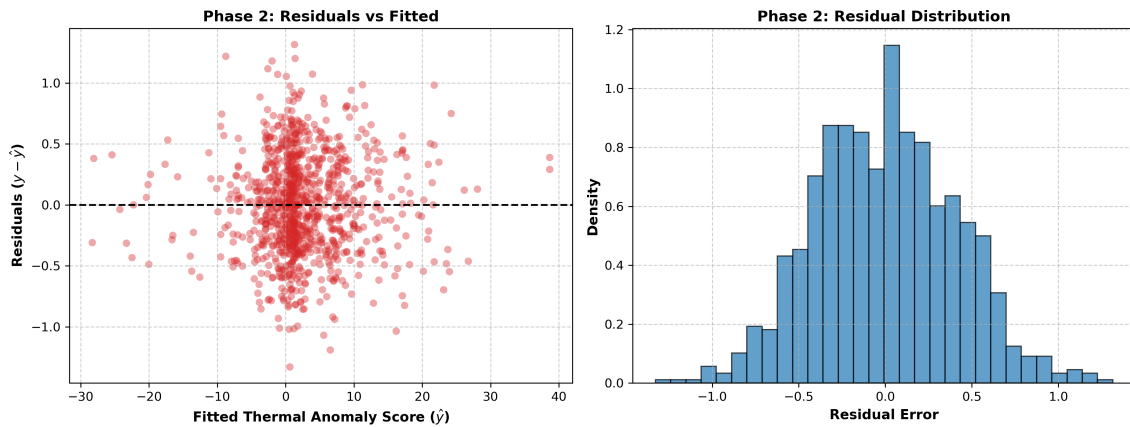


Figure 4: Residual analysis for the final degree-10 Ridge model for `var2`.

5 Inference and Prediction

After selecting the polynomial degree and training the final models, the trained models are used to make predictions on the corresponding test datasets.

The inference process is:

$$X_{\text{test}} \longrightarrow \text{Polynomial Feature Expansion} \longrightarrow \text{Trained Regression Model} \longrightarrow \hat{y}_{\text{test}}.$$

The actual target values of the test datasets are hidden. Therefore, test-set MSE and R^2 cannot be calculated locally.

The final prediction files are:

- BT2024182_pred_var1.csv
- BT2024182_pred_var2.csv

Each file contains the predicted values of y for its corresponding test dataset.

6 Implementation

The implementation is divided into separate modules for polynomial feature generation, regression models, cross-validation, evaluation, training, and inference.

The main implementation files are:

- `polynomial.py`: polynomial feature generation.
- `models.py`: regression models and regularization.
- `cross_val.py`: cross-validation and degree selection.
- `evaluate.py`: evaluation utilities.
- `train_var1.py`: training pipeline for `var1`.
- `train_var2.py`: training pipeline for `var2`.
- `inference.py`: generation of predictions on the test datasets.

The training scripts perform model selection and subsequently train the final models using the complete training datasets. The inference script uses the trained models to generate predictions for the test datasets.

7 Conclusion

Polynomial regression was applied to both personalized datasets.

For `var1`, polynomial degree 5 was selected. Regularization and feature selection were used to control the large polynomial feature space. The final model achieved a training MSE of approximately 0.24041 and a training R^2 of 0.97539.

For `var2`, several polynomial degrees were evaluated using 5-fold cross-validation. Degree 10 achieved the lowest validation MSE among the evaluated degrees, with a CV MSE of 0.24532 and a CV R^2 of 0.99462. Therefore, degree 10 with Ridge regularization ($\alpha = 0.05$) was selected as the final model.

The trained models were subsequently used for inference on the hidden test datasets, producing the two required prediction CSV files.